



Performance Testing

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Testing
Type of Testing	Load Testing
Target Module / API	HDI Prediction
Test Environment	Local System
Test Date	

Step 2: Test Scenarios

S. No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits HDI prediction request	1	30	Prediction result is displayed correctly.
2	Multiple prediction requests	10	60	Application responds successfully without errors.
3	Continuous prediction requests	20	120	Application remains stable and returns accurate predictions.

Step 3: Performance Test Results

S. No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Average)	< 2 seconds	1.2 seconds	Pass	Fast response for prediction requests.
2	Response Time (Max)	< 5 seconds	2.8 seconds	Pass	Maximum response time is within the limit.
3	Throughput (Req/sec)	10 Req/sec	12 Req/sec	Pass	System handled requests efficiently.
4	Error Rate	< 1%	0%	Pass	No errors observed during testing.
5	CPU Utilization	< 80%	45%	Pass	CPU usage remained stable.
6	Memory Utilization	< 80%	52%	Pass	Memory usage was within acceptable limits.

Step 4: Observations & Analysis

Key Findings:

The HDI Prediction system performed efficiently during testing. The average responses time was below 2 seconds, CPU and memory usage remained within acceptable limits, and all prediction requests were processed successfully without failures.

Bottlenecks Identified:

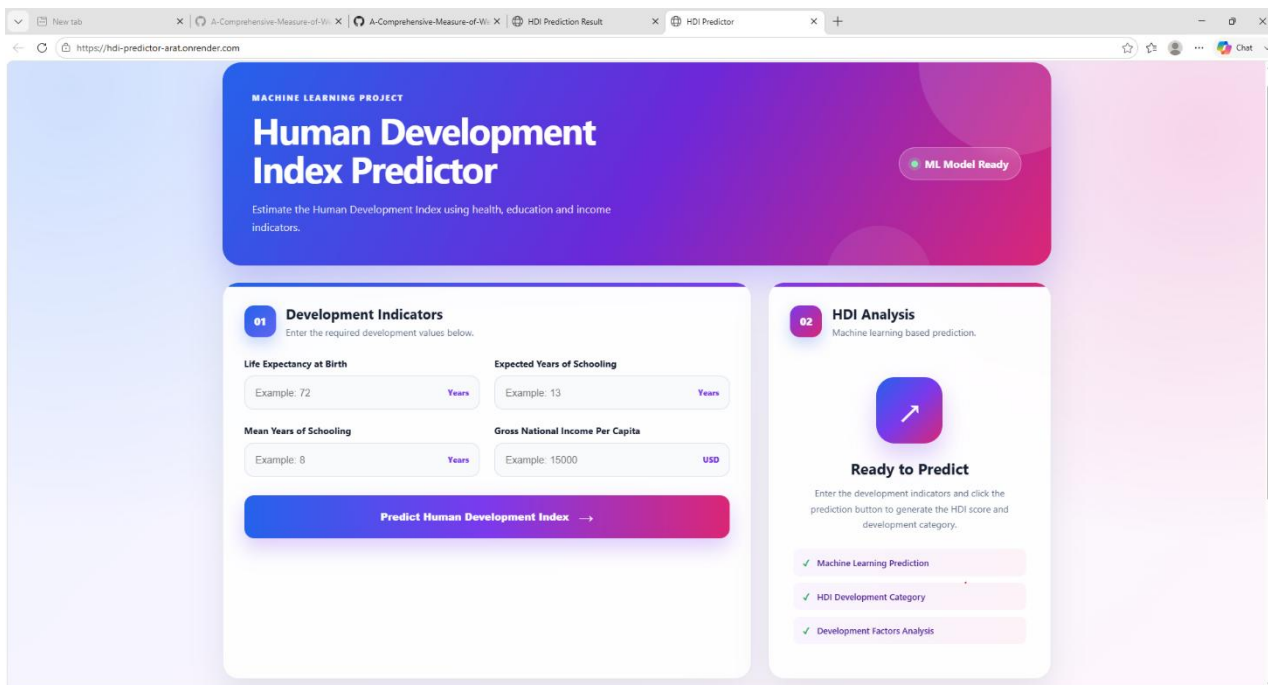
No major performance bottlenecks were identified during testing. The application remained stable under the tested workload with no significant delays or errors.

Optimization Steps taken:

Input validation was implemented to prevent invalid data, unnecessary computations were minimized, and the trained machine learning model was loaded efficiently to improve prediction speed and overall application performance.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



A screenshot of a web browser displaying the 'HDI Predictor' application. The browser's address bar shows the URL 'https://hdi-predictor-arat.onrender.com/predict'. The page has a light blue header with the title 'HDI Predictor' and a subtitle 'Human Development Analysis using ML'. Below the header, a green checkmark indicates 'Prediction Completed Successfully!'. The main content area features a 'HDI Score' of 0.735 out of 1.000, categorized as 'High Human Development'. A 'Development Category' section shows a bar chart with three bars: 'High Human Development' (blue, 0.735), 'Medium Human Development' (orange, 0.550), and 'Low Human Development' (green, 0.299). The 'HDI Scale Analysis' section provides a range of development levels: Low (Below 0.550), Medium (0.550 - 0.699), High (0.700 - 0.799), and Very High (0.800+). The 'Development Factors Analysis' section lists Life Expectancy (73.0 years), Expected Years of Schooling (14.0 years), Mean Years of Schooling (9.0 years), and GNI Per Capita (\$16000.0). A 'Make New Prediction' button is located at the bottom left.